

INDUSTRIAL PRODUCT DATASHEET

reComputer Super J401 & optional accessories

The reComputer Super J401 delivers up to 157 TOPS, dual GbE, LTE, and 4x MIPI-CSI for seamless AI deployment in robotics, vision AI, and generative AI applications.

SKU

E2025032601

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Overview



The reComputer Super J401 is an edge AI computer delivering up to 157 TOPS in MAXN Super Mode, with configurable power modes from 10W to 40W and robust thermal design for operation from -20°C to 60°C. It features rich interfaces including dual RJ45, SIM card, 4x USB 3.2, HDMI 2.1, CAN, M.2 Key E/M, Mini-PCIe, and 4x MIPI-CSI, and comes pre-installed with JetPack 6.2, a 128GB NVMe SSD, and AI framework support.

Key Features

Note:

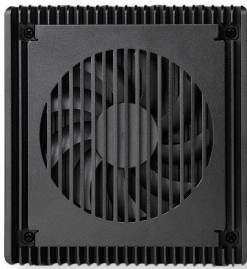
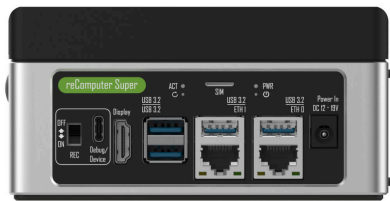
For Orin Nano module recommends to use 12V power adapter.

For Orin NX module recommends to use 19V power adapter.

- **Super Boost AI Performance:** Delivers up to 157 TOPS in MAXN Super Mode, ideal for demanding applications like vision AI, robotics, and generative AI.
- **Energy-Efficient Operation:** Configurable power modes from 10W to 40W allow for optimized performance and energy consumption.
- **Robust Thermal Design:** Operates reliably in temperatures from -20°C to 60°C at 40W, and up to 65°C at 25W.
- **Rich interfaces:** Features 2x RJ45, SIM card, 4x USB 3.2, HDMI 2.1 and CAN. Also including M.2 Key E/M, Mini-PCIe, 4x Mipi-CSI for connectivity and cameras.
- **Ready for Deployment:** Pre-installed with JetPack 6.2 ([Supports JP7.2](#)) and a 128GB NVMe SSD, supporting AI frameworks like NVIDIA Isaac, Hugging Face, and ROS2/1, enabling rapid development of various AI applications.
- Memory Bandwidth Upgrade
- Higher Throughput
- Improved higher frame rate processing
- Enhanced Power Management
- Reduced Inference Latency

Built for developers, the reComputer Super J401 is an enhanced upgrade over the [reComputer J401](#). It takes edge AI development further with key enhancements: dual GbE (RJ45) ports, LTE, 4x Mipi-CSI for Camera, and full support for MAXN Mode.

Product Gallery





Specification

	reComputer Super J3010	reComputer Super J3011	reComputer Super J4011
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reComputer Super J4012

Applications Processor(AP)	NVIDIA Orin™ Nano 4GB	NVIDIA Orin™ Nano 8GB	NVIDIA Orin™ NX 8GB	N V I D I A O r i n ™ N X 1 6 G B
AI Performance	Orin Nano 4GB – 34 TOPS(MAXN)	Orin Nano 8GB – 67 TOPS(MAXN)	Orin NX 8GB – 117 TOPS(MAXN)	O r i n N X 1 6 G B – 1 1 7 T O P S (M A X N)
GPU	512 CUDA Cores 16 Tensor Cores	1024 CUDA Cores 32 Tensor Cores		
GPU Max Frequency	1020 MHz (MAXN)		1173 MHz (MAXN)	

CPU	6-core Arm® Cortex®-A78AE		6-core NVIDIA Arm® Cortex A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3		8 - core NVIDIA Arm® Cortex A78AE v8.2 64-bit CPU 2MB L2 + 4MB L3
CPU Max Frequency	1.7 GHz (MAXN_SUPER)		2.0 GHz		
DL Accelerator	-	-	1 x NVDLA v2.0	2 x NVDLA v2.0	
DL Max Frequency	-	-	1.23 GHz		
Vision Accelerator	-	-	1 x PVA v2.0		

Memory	4GB 64-bit LPDDR5 51 GB/s (MAXN)	8GB 128-bit LPDDR5 102 GB/s (MAXN)	8GB 128-bit LPDDR5 102.4GB/s (MAXN)	1 6 G B 1 2 8 - b i t L P D D R 5 1 0 2 . 4 G B / s (M A X N)
Power	7W - 10W - 25W	10W - 15W - 25W	10W - 15W - 25W - 40W	1 0 W - 1 5 W - 2 5 W - 4 0 W
Video Encode	1080p30 supported by 1-2 CPU cores		1x 4K60 (H.265) 3x 4K30 (H.265) 6x 1080p60 (H.265) 12x 1080p30 (H.265)	
Video Decode	1x 4K60 (H.265) 2x 4K30 (H.265) 5x 1080p60 (H.265) 11x 1080p30 (H.265)		1x 8K30 (H.265) 2x 4K60 (H.265) 4x 4K30 (H.265) 9x 1080p60 (H.265) 18x 1080p30 (H.265)	
CSI camera	Up to 4 cameras (8 via virtual channels***) 8 lanes MIPI CSI-2 D-PHY 2.1 (up to 20Gbps)		Up to 4 cameras (8 via virtual channels***) 8 lanes MIPI CSI-2 D-PHY 2.1 (up to 20Gbps)	
Mechanical	69.6mm x 45mm 260-pin SO-DIMM connector			

Carrier Board		
Storage	1x M.2 KEY M PCIe (M.2 NVMe 2280 SSD 128G included)	
Networking	M.2 KEY E	1x M.2 Key E for WiFi/Bluetooth module
	Mini PCIe	1x mini-PCIe for LTE 4G module
	Ethernet	2x RJ45 Gigabit Ethernet

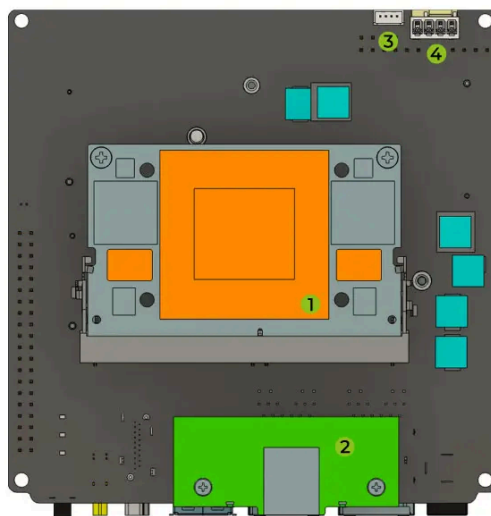
I/O	USB	4x USB 3.2 Type-A (5Gbps); 1x USB 2.0 Type-C (Device Mode/Debug);
	Camera	4x mipi CSI(2-lane 15-Pin)
	CAN	1 x CAN(4-Pin Header)
	Display	1x HDMI 2.1
	Fan	1x 4-Pin 1.25mm Fan Connector (5V PWM) 1x 4-Pin 2.54mm Fan Connector (12V PWM)
	Extension Port	1x 40-Pin extension header 1x 12-Pin control and UART header
	RTC	1x RTC 2-pin 1x RTC Socket
	LED	2x LED(PWR and ACT)
	Button	1x PWR 1x RESET
	Switch	1x REC
	Antenna hole	4x Antenna Hole
Power	12-19V 5525 Barrel DC Jack	
Jetpack Version	Jetpack 6.2	
Mechanical	Dimensions (W x D x H)	130mm x 120mm x 66mm
	Weight	1110g
	Installation	Desk, Wall-mounting
Operating Temperature	-20°C~65°C(25W Mode) -20°C~60°C(MAXN Mode)	
Warranty	2 Years	
Certification	RoHS,REACH,CE,FCC,UKCA, KC	

Hardware Overview



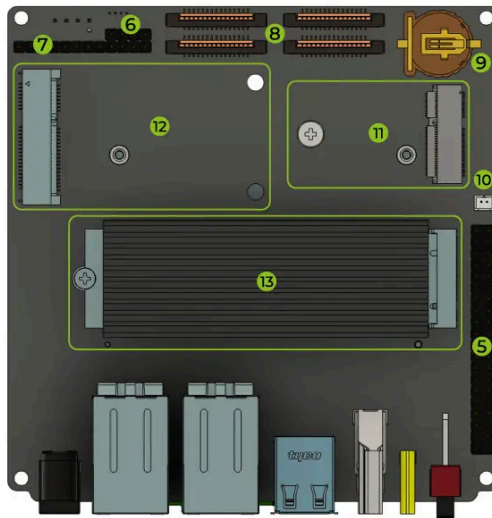
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|------------------|---------------------|-------------------------|
| ① DIP Switch | ④ 4x Type A USB 3.2 | ⑦ SIM Card Slot |
| ② Type C USB 2.0 | ⑤ 2x Pinhole Button | ⑧ 2x RJ45 Ethernet Port |
| ③ HDMI 2.1 | ⑥ 2x LED Indicator | ⑨ Barrel DC Jack 5525 |

Top View



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|---------------------------------|
| ① Nvidia Jetson Orin Module |
| ② Extension Board |
| ③ 4-Pin Fan Connector (5V PWM) |
| ④ 4-Pin Fan Connector (12V PWM) |

Bottom View



- 5 40-Pin Extension Header
- 6 CAN Header
- 7 12-Pin control and UART header
- 8 4x 15-Pin MIPI CSI
- 9 RTC Battery Socket(CR1220)
- 10 RTC 2-Pin Header(3V)
- 11 M.2 Key E Slot
- 12 Mini PCIe Slot
- 13 M.2 Key M Slot(with SSD)

Applications

- **AI Workstation for Beginners:** Use reComputer Super AI box as your first AI workstation, diving into NVIDIA Jetson for Beginners (<https://github.com/Seeed-Projects/reComputer-Jetson-for-Beginners>) for a fast track to vision AI, Generative AI, and robotics. Learn from zero to real projects: master Linux basics on Jetson, build an end-to-end AI NVR/AMR, and explore CV, GenAI, ROS, agentic AI, and TensorRT optimization with tailored hardware integration.
- **One-Click AI Deployment with reComputer AI Lab:** reComputer AI Lab (<https://sensecraft.seeed.cc/ai-lab>) provides one-click tools and ready-to-use resources including optimized edge AI model demos for CV, LLM, VLM, interactive tutorials, deployment tools, containerized applications, and community projects to speed up Jetson development.
- **Seamless Setup with Jetson Develop Tool:** Seeed Jetson Develop Tool (<https://github.com/Seeed-Projects/Seeed-Jetson-DevelopTool>) is a desktop client that unifies firmware flashing, first-boot initialization, device diagnostics, remote development, app deployment, skill-based automation, OTA updates, and community resources in one GUI for easy reComputer Jetson setup.
- **One-Line Deployable AI Project Examples:** Jetson-examples (<https://github.com/Seeed-Projects/jetson-examples>) offers one-line deployable projects for generative AI large models like LeRobot, Ollama, Qwen, and computer vision such as Ultralytics YOLO, with pre-configured environments for single-command verification of model performance on your GPU.
- **Multi-Camera 360° Vision AI for Autonomous Operation:** Supports up to 4 USB cameras (<https://www.seeedstudio.com/X10-USB-wired-camera-p-6506.html>) or MIPI-CSI cameras (<https://www.seeedstudio.com/IMX219-77-Camera-77-FOV-Applicable-for-Jetson-Nano-p-4608.html>) to create a panoramic vision network. Leverages spatial foundation models for multi-stream real-time depth perception, environmental awareness, and blind-spot elimination for safer autonomous operation.

Part List

Jetson Orin™ NX 16GB/NX 8GB /Nano 8GB/Nano 4GB module	x 1
Seeed Carrier Board (reComputer Super J401)	x 1
128GB NVMe SSD	x 1
Aluminum Case and Heatsink with Fan	x 1
USB Cable; Type A to Type C	x 1
User Manual	x 1

Resource Links

- <https://www.seeedstudio.com/reComputer-Super-Bundle.html>
- [User Manual&Datasheet](#)
- [Temperature Test Report](#)
- [Schematic](#)
- [3D File](#)
- [Mechanical Document-reComputer Super](#)
- [Mechanical Document-reComputer Super PCBA](#)
- [Seeed Nvidia Jetson Success Cases](#)
- [Seeed Jetson One Pager](#)

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